AMERICAN UNIVERSITY OF PHNOM PENH

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COSC 251 Fall 2026: Assignment 3 - Linked List (50 points)
Deadline: Oct 21 11:59 pm
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Overview: For this assignment, you are to implement the *LinkedList* class, the *DoublyLinkedList* class, and change your *Stack* class and *Queue* class from Assignment 2 to use your *LinkedList* class.

Consider the time complexity when implementing each method to get full points for this assignment. Your *LinkedList* class and the *DoublyLinkedList* class must make use of generics, and follow the principles and functionalities we discussed in class. Your *Stack* class and *Queue* must follow all requirement set in assignment 2 but make use of your *LinkedList* class instead of Array.

Project Requirements:

1. The *LinkedList* class must make use of the *Link* class and must have the following constructors and methods:
   (a) `public LinkedList()`
   (b) `public void insertFirst(T newItem)`
   (c) `public void insertLast(T newItem)`
   (d) `public void insertAt(T newItem)`
   (e) `public boolean deleteFirst()`
   (f) `public boolean deleteLast()`
   (g) `public boolean deleteAt(int index)`
   (h) `public boolean deleteKey(T key)`
   (i) `public int find(T key)`
   (j) `public boolean isEmpty()`
   (k) `public String toString()`
   (l) `public void display()`
2. The *Link* class must have the following attributes:
   (a) `T data`
   (b) `Link next`
3. The *DoublyLinkedList* class must make use of the *DoublyLink* class and must have the following constructors and methods:
   (a) `public LinkedList()`
   (b) `public void insertFirst(T newItem)`
   (c) `public void insertLast(T newItem)`
   (d) `public void insertAt(T newItem)`
   (e) `public boolean deleteFirst()`
   (f) `public boolean deleteLast()`

(g) `public boolean deleteAt(int index)`

(h) `public boolean deleteKey(T key)`

(i) `public int find(T key)`

(j) `public boolean isEmpty()`

(k) `public String toString()`

(l) `public void displayForward()`

(m) `public void displayBackward()`

4. The *DoublyLink* class must have the following attributes:

   (a) `T data`

   (b) `Link next`

   (c) `Link previous`

5. The *LinkedList* class and the *DoublyLinkedList* class must be double-ended. Meaning they must have head and tail pointers.

6. The *LinkedList* class and the *DoublyLinkedList* class must be generic classes.

7. There must be the *Link* class for the *LinkedList* class and the *DoublyLink* class for the *DoublyLinkedList* class. Both the *Link* class and the *DoublyLink* class must be generic classes.

8. Change the implementation of your *Stack* class from assignment 2 to make use of the *LinkedList* class above. All other requirements from assignment 2 still applies to your *Stack* class.

9. Change the implementation of your *Queue* class from assignment 2 to make use of the *LinkedList* class above. All other requirements from assignment 2 still applies to your *Queue* class.

10. There must be a comment on all your methods to document its time complexity.

What to turn-in:

This is a group assignment, each group should only have submission.

1. Your *LinkedList* class with the required methods and functionalities. The file must be named LinkedList.java and the class must be named LinkedList.
2. Your *Link* class with the required methods and functionalities. The file must be named Link.java and the class must be named Link.
3. Your *DoublyLinkedList* class with the required methods and functionalities. The file must be named DoublyLinkedList.java and the class must be named DoublyLinkedList.
4. Your *DoublyLink* class with the required methods and functionalities. The file must be named DoublyLink.java and the class must be named DoublyLink.
5. Your *Stack* class with the required methods and functionalities. The file must be named Stack.java and the class must be named Stack.
6. Your *Queue* class with the required methods and functionalities. The file must be named Queue.java and the class must be named Queue.
7. Optional: any other supporting Java classes.
8. Note: Do not submit your Main Java class; your Java classes that you submit must not have any main method.

Grading:

1. Program Design (10): How well your solution is thought out.
2. Program Execution (15): The program runs correctly.
3. Specification Satisfaction (15): The program satisfies specifications completely and correctly.
4. Coding style and readability (10): Well-formatted, understandable code, and meaningful comments.